Written By: Tony Cruise

First Published: Micro’s Gazette – Issue 002 (March/April 1989)

This issue I will start to cover some of the routines that will be used in our extended Basic. We will start off with the new sprite commands.

You can have up to 32 sprites on the screen at once but if you want to move more than about five or six, your program slows down considerably. The routines this issue allow you to specify X and Y velocities for each of the 32 sprites and then they will be moved independently of your Basic program.

I have included an assembler listing explaining how it works as well as a basic loader program for those people who do not have assemblers and a demonstration program.

Machine Code Program

		;	Program to move sprites by velocity	
		;		
D000		ORIGIN	ORG D000H	
		;		
		;	Label settings	
		;		
		HTIMI	EQU FD9FH	; Hook jump – Timer
		SCRMOD	EQU FCAFH	; RAM storage – Screen mode
		SETWRT	EQU 0053H	; ROM routine – Set video write
		SETRD	EQU 0050H	; ROM routine – Set video read
		;		
D000	3EC3	START	LD A,C3H	; Value for JUMP instruction
D002	329FFD		LD (HTIMI),A	; Load into Hook
D005	210CD0		LD HL,QUE	; Address to jump to
D008	22A0FD		LD (HTIMI+1),HL	; Load into Hook
D00B	C9		RET	;
		;		
D00C	CD14D0	QUE	CALL SPRMOV	; Call SPRMOV routine
D00F	F7879C77		DEFB F7H,87H,9CH,77H	; Disk delay routine, remove if

D013	C9	20	RET	; up do not have a disk drive
		21 ;		
D014	3AAFFC	22 SPRMOV	LD A,(SCRMOD)	; Get screen mode
D017	FE00	23	CP 0	;
D019	C8	24	RET Z	; Make sure not screen 0
D01A	3A47D1	25	LD A,(MOVFLG)	;
D01D	FE00	26	CP 0	; Get move flag
D01F	C8	27	RET Z	;
D020	21001B	28	LD HL,1B00H	; Return if move set to off
D023	1187D0	29	LD DE,SPRTBL	;
D026	018000	30	LD BC,128	;
D029	CD78D0	31	CALL LDIRVM	; Get sprite table
D02C	2187D0	32	LD HL,SPRTBL	; Set pointer to sprite table
D02F	0620	33	LD B,32	; Set loop counter
D031	1107D1	34	LD DE,SPRVEL	; Set point to velocities
D034	7E	35 LOOP1	LD A,(HL)	;
D035	FED1	36	CP 209	; Is the sprite on screen?
D037	281A	37	JR Z,LOOP2	; No skip
D039	4F	38	LD C,A	;
D03A	1A	39	LD A,(DE)	; Get Y velocity
D03B	81	40	ADD A,C	; Add Y velocity
D03C	FEDC	41	CP 220	; Off Top of screen
D03E	3804	42	JR C,LOOP3	; No skip
D040	3EC0	43	LD A,192	; Yes, correct value
D042	1805	44	JR LOOP4	;
D044	FEC0	45 LOOP3	CP 192	; Off bottom of screen?
D046	3801	46	JR C,LOOP4	; No skip
D048	AF	47	XOR A	; Yes, correct value
D049	77	48 LOOP4	LD (HL),A	; Store new value
D04A	23	49	INC HL	;
D04B	13	50	INC DE	; Increment pointers
D04C	7E	51	LD A,(HL)	;
D04D	4F	52	LD C,A	;
D04E	1A	53	LD A,(DE)	; Get X velocity
D04F	81	54	ADD A,C	; Add X velocity
D050	77	55	LD (HL),A	; Store new value
D051	2B	56	DEC HL	;
D052	1B	57	DEC DE	; Restore pointers
D053	13	58 LOOP2	INC DE	;
D054	13	59	INC DE	;
D055	23	60	INC HL	;
D056	23	61	INC HL	;
D057	23	62	INC HL	;
D058	23	63	INC HL	; Update pointers
D059	05	64	DEC B	; Decrease loop counter
D05A	20D8	65	JR NZ,LOOP1	; Next loop

D05C	21001B	66	LD HL,1B00H	;
D05F	1187D0	67	LD DE,SPRTBL	;
D062	018000	68	LD BC,128	;
D065	CD69D0	69	CALL LDIRMV	; Save sprite table
D068	C9	70	RET	;
		71 ;		
		72 ;	Move a block of memory from video RAM	
		73 ;		
D069	CD5300	74 LDIRMV	CALL SETWRT	
D06C	1A	75 LP1	LD A,(DE)	
D06D	D398	76	OUT (98H),A	
D06F	13	77	INC DE	
D070	0B	78	DEC BC	
D071	78	79	LD A,B	
D072	B1	80	OR C	
D073	FE00	81	CP 0	
D075	20F5	82	JR NZ,LP1	
D077	C9	83	RET	
		84 ;		
		85 ;	Move a block of memory to video RAM	
		86 ;		
D078	CD5000	87 LDIRVM	CALL SETRD	
D07B	DB98	88 LP2	IN A,(98H)	
D07D	12	89	LD (DE),A	
D07E	13	90	INC DE	
D07F	0B	91	DEC BC	
D080	78	92	LD A,B	
D081	B1	93	OR C	
D082	FE00	94	CP 0	
D084	20F5	95	JR NZ,LP2	
D086	C9	96	RET	
		97 ;		
		98 ;	Data storage	
		99 ;		
D087		100 SPRTBL	DEFS 128	
D107		101 SPRVEL	DEFS 64	
D147	00	102 MOVFLG	DEFB 0	
		103 END		

Basic Loader

10 CLS:CLEAR 200,&HCFFF:DEFINTA-Z:A=&HD00020 READ A$:IF A$ <> “@” THEN POKE A, VAL(“&H”+A$):A=A+1:GOTO 20 30
POKE &HD147,0
40 PRINT”

INSERT TAPE/DISK TO SAVE PROGRAM”

50 PRINT”

AND PRESS ANY KEY”
60 A$=INPUT$(1):PRINT:PRINT “

SAVING ....”
70 BSAVE”SPRITE.OBJ”,&HD000,&HD147
80 END
100 DATA 3E,C3,32,9F,FD,21,0C,D0,22,A0,FD,C9,CD,14,D0
110 DATA F7,87,9C,77 (or DATA 00,00,00,00 if you don’t have a disk drive)
120 DATA C9,3A,AF,FC,FE,00,C8,3A,47,D1,FE,00,C8,21,00
130 DATA 1B,11,87,D0,01,80,00,CD,78,D0,21,87,D0,06,20
140 DATA 11,07,D1,7E,FE,D1,28,1A,4F,1A,81,FE,DC,38,04
150 DATA 3E,C0,18,05,FE,C0,38,01,AF,77,23,13,7E,4F,1A
160 DATA 81,77,2B,1B,13,13,23,23,23,23,05,20,D8,21,00
170 DATA 1B,11,87,D0,01,80,00,CD,69,D0,C9,CD,53,00,1A
180 DATA D3,98,13,0B,78,B1,FE,00,20,F5,C9,CD,50,00,DB
190 DATA 98,12,13,0B,78,B1,FE,00,20,F5,C9,@

Basic Program Example

10 COLOR 15,1,9:SCREEN 2,2:SPRITE$(0) = STRING$(32,244):DEF FNA(X) = INT(RND(1)*X)+1:A = RND(-TIME) 20 STOP ON:ON STOP GOSUB 11030 POKE &HD147,1:FOR A=0 TO 31
40 PUT SPRITE A,(128,96),FNA(15),0
50 XV=FNA(5)-3:IF XV<0 THEN XV=XV+256
60 YV=FNA(5)-3:IF YV<0 THEN YV=YV+256
70 IF XV+YV=0 THEN 50
80 POKE &HD107+A*2,YV:POKE &HD108+A*2,XV:NEXT
90 IF NOT(STRIG(0)) THEN 90 ELSE POKE &HD147,0
100 IF NOT(STRIG(0)) THEN 100 ELSE POLE &HD147,1:GOTO 80
110 POKE &HD147,0:END

Type in the machine code loader first and save it to disk or tape. Now type in the Basic example program and save it to tape or disk.

Then use BLOAD”SPRITE.OBJ”,R to run the machine code program and then RUN the example Basic program.

Next month I will cover the collision testing routines. Bye for now!